

Yukta Seelam

yuktas@bu.edu | (857) 540-9665 | Boston, MA (Open to relocation)

SUMMARY

Graduate student in Robotics and Autonomous Systems at Boston University specializing in machine learning, deep learning, and data-driven control. Experienced in building and deploying ML models for perception, reinforcement learning, and generative applications using PyTorch, TensorFlow, and AWS.

ACADEMIC PROFILE

Boston University **3.8/4.0**
Masters in Robotics and Autonomous Systems

Expected Jan 2026

Indian Institute of Technology Kanpur
B. Tech in Electrical Engineering

May 2024

EXPERIENCE & RESEARCH

Hybrid Local-Cloud RL for Autonomous Driving in CARLA – Red Hat / Boston University

May 2025 – Sep 2025

- Developed a distributed reinforcement learning framework (UniLCD) enabling dynamic policy switching between local and cloud models to balance latency and energy efficiency.
- Trained PPO-based agents in CARLA with adaptive inference pipelines and confidence-driven offloading.
- Implemented real-time video logging and rollout evaluation using OpenCV, NumPy, and custom Python utilities.

Samsung Research & Development – India

May 2023 – Aug 2023

- Deployed deep learning models (YOLOv5, RefineDet) for object detection in robotic vacuum navigation pipelines.
- Enhanced cross-platform model compatibility using TensorRT and ONNX, reducing inference latency by 25%.
- Built AWS Lambda-based automation for IP whitelisting, improving operational reliability and data security.

PROJECTS

Music Synthesis using Generative Machine Learning - IIT Kanpur

Jan 2024 – May 2024

- Designed a Variational Autoencoder (VAE) for pitch-contour generation, adapting Western synthesis methods for Indian classical music.
- Developed a unique note-transition preservation module, improving reconstructed pitch continuity by 40%.

Mobile Charging Station for EVs - IIT Kanpur

Aug 2023 – Dec 2023

- Applied federated learning with RNN and LSTM models to predict optimal charging station placement.
- Proposed and implemented transformer-based temporal forecasting that outperformed RNN baselines by 13% RMSE reduction.

Audio Event Detection - IIT Kanpur

Aug 2023 – Dec 2023

- Designed a Variational Autoencoder (VAE) for pitch-contour generation, adapting Western synthesis methods for Indian classical music.
- Developed a unique note-transition preservation module, improving reconstructed pitch continuity by 40%.

AWARDS AND PUBLICATIONS

- 3rd Place – IEEE ICRA DJI RoboMaster AI Challenge 2022 (only Indian team internationally)
- Paper: “Towards Autonomous Shooting Rover via Situation-Aware Visual Perception and Dynamic Action Execution”, IEEE/SICE SII 2024 (Vietnam)
- Poster: “Design of AI Ground Robots for Intelligent Combat in a Dynamic Arena”, ICRA 2022

LEADERSHIP POSITIONS

- Team Head – Robotics Club - IITK | President – Toastmasters IIT Kanpur | Captain – IITK Basketball Team | Student Ambassador – BU Robotics

SKILLS & COURSEWORK

- Programming:** Python, C++, MATLAB, Bash, SQL
- ML/DL Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, NumPy, Pandas
- Data & Deployment:** AWS (Lambda, S3), TensorRT, OpenCV, Matplotlib, Seaborn
- Specializations:** Deep Learning, Reinforcement Learning, Federated Learning, Generative Models, Computer Vision